

Variable DC Power Supplies

Understanding the complicated technical specifications in datasheets can help you select the right power supply for your range of applications



ANKIT GUPTA

Variable power supply is generally the first equipment to find its place on an electronics engineer's desk. It is undoubtedly the most popular test tool after a multimeter. There are a wide variety of power supplies available. Selecting the right one for your project requires a good understanding of how they are specified.

There are basically two types of power supplies that are used as test equipment: Linear power supplies and switch-mode power supplies. Switch-mode power supplies are lighter and have replaced linear power supplies in many applications. However, linear power supplies continue to be preferred because of their accuracy, durability and very little noise at the

output. These power supplies are excellent for applications that are sensitive to small power variations.

Each manufacturer has a number of power supplies that are majorly specified by output voltage and current rating. The ratings that you would require will completely depend on the application area. For example, for regular electronics testing, a 32V/2A power supply is good enough. But if you are working on a project for Indian railways, you need at least 110V/5A because input supply in Indian railways is 110V.

Specifications in datasheets look complicated at first sight but these can be easily understood by categorising them. Broadly, there are five categories: Accuracy, resolution, regulation, AC characteristics and packaging. In all the organisations, equipment are regularly calibrated so that they keep

delivering these specs without any deviation.

Accuracy

Accuracy is one of the most critical requirements for a power supply. It indicates the degree of closeness of the

Power Supplies

Linear power supplies. In linear power supplies, the input AC voltage is rectified to create DC, which is further filtered and regulated to produce user-selectable voltage. Linear supplies are heavier because these use a 50Hz or 60Hz transformer and associated filters.

SMPS power supplies. SMPS-based power supplies work in the same way as linear power supplies but reduce the size of the transformer by chopping the AC line with high frequency. With high frequency, the transformer size reduces significantly.